Fracture Mechanics 2026 · Homework · Group parameters

The DCDC test

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Take the row of your group number and write the pair on your cover sheet and on the first line of your report. No two groups have the same pair, and the answers to Questions 3 and 5 depend on it: W/R fixes the stress concentration k_t and therefore the whole curve $\hat{g}(a/R)$, and R decides whether the toughness criterion governs anywhere, so whether the test can be run at all on the concrete of Question 5.

group	W/R	R (mm)
1	2.0	150
2	2.0	1000
3	2.5	400
4	2.5	150
5	3.0	1000
6	3.0	400
7	3.5	150
8	3.5	1000
9	4.0	400
10	4.0	1000
11	2.0	400
12	2.5	1000
13	3.0	150
14	3.5	400
15	4.0	150

Everything else is common to all groups: the geometry of the subject, plane strain, $\nu = 0.2$, and the length $L \geq R + a + 3W$. Group numbers, the folder for the submission and the deadline are announced separately.